

The CHLA Insulin Calculator will accurately determine the correct dose of rapid-acting insulin to be administered based on a specific patient order, the grams of carbohydrates eaten, and the measured blood glucose.

Insulin/Carbohydrate Ratio (ICR):

- 1 unit of insulin covers **X** grams of carbohydrates eaten
- Allowed range: 1 – 100 gm:1 unit
 - “Usual” ICR (~10th – 90th %ile) varies by age
 - 0 - 5y: 10 – 50
 - 6 - 11y: 5 – 30
 - ≥ 12y: 4 – 12

✓ ordered value; click to confirm → ✓

⚠ indicates an extremely low or high value that should be clarified

- 0 - 5y: LOW: < 10 | HIGH: > 50
- 6 - 11y: LOW: < 5 | HIGH: > 30
- ≥ 12y: LOW: < 4 | HIGH: > 12

Correction Factor(CF) -- also called Insulin Sensitivity Factor (ISF):

- 1 unit of insulin reduces a Blood Glucose elevated over Target Blood Glucose by **X** mg/dL
- Allowed range: 4 – 300 mg/dL:1 unit
 - “Usual” CF/ISF (~10th – 90th %ile) varies by age
 - 0 - 5y: 50 – 200
 - 6 - 11y: 25 – 100
 - ≥ 12y: 15 – 50

✓ ordered value; click to confirm → ✓

⚠ indicates an extremely low or high value that should be clarified

- 0 - 5y: LOW: < 40 | HIGH: > 200
- 6 - 11y: LOW: < 15 | HIGH: > 200
- ≥ 12y: LOW: < 8 | HIGH: > 120

Target Blood Glucose:

- blood glucose target above which correction must be applied
- Age, individual and context-dependent
- Allowed range: 100 – 250 mg/dL

- “Usual” Target BG (~10th – 90th %ile):

- 0 - 5y: 150 – 200
- 6 - 11y: 130 – 180
- ≥ 12y: 120 – 150

✔ ordered value; click to confirm → ✔

⚠ indicates an extremely low or high value that should be clarified

- 0 - 5y: LOW: < 130 | HIGH: > 250
- 6 - 11y: LOW: < 120 | HIGH: > 200
- ≥ 12y: LOW: < 120 | HIGH: > 200

****The calculator is not enabled for input of carbs eaten or measured blood glucose until all above parameters are confirmed by clicking the ✔ or ⚠***

Carbohydrates Eaten:

- Enter the grams of carbohydrates eaten
- calculator will accept any value from 0 – 300 gm
- ⚠ carbs > 250 gm; verify and correct if data entry was incorrect
 - Carb coverage will only be provide up to 250 gms
- ✖ unacceptable entry which must be corrected

Blood Glucose for Calculation:

- Enter the current glucose measurement
- calculator will accept any value from 1 – 2,000 mg/dL
- ⚠ BG > 600 mg/dL; verify and correct if data entry was incorrect
 - Correction is only performed for BG up to 600
- 🚰 BG > 70 mg/dL: hypoglycemia
 - STOP – Calculator will not calculate dose
 - DO NOT GIVE INSULIN
 - requires rescue intervention for low BG
- ✖ unacceptable entry which must be corrected

To complete calculation of dose to administer:

- Assure all parameters and patient values are valid
- Click button
- Dose will be displayed, **rounded** per order increments, here:

Calculated Dose to Administer

→ units

- Transfer calculated dose to MAR after independent 2nd witness check and dose has been administered
- **REMEMBER**
 - *Change time from default to time actually administered*
 - *Change dose from “**1 dose**” to “**X units**” actually administered*